

Arpon Roy

Kakinada, Andhra Pradesh, India

royarpon007@gmail.com | 8657967725 | LinkedIn | GitHub | Portfolio

Summary

Computer Science undergraduate with a strong record of self-driven, production-grade software development across the MERN stack, Java, and Spring Boot. Skilled in designing secure REST APIs, scalable system architecture, and real-time data workflows, with hands-on exposure to Generative AI, Docker, AWS, and CI/CD pipelines. Demonstrated initiative through multiple independently built full-stack platforms and two technical internships. Seeking a scholarship opportunity to further pursue advanced study and research in Computer Science and Software Engineering.

Education

Jawaharlal Nehru Technological University Kakinada (JNTUK)

Sep 2023 – Apr 2027

B.Tech in Computer Science and Engineering

CGPA: 8.02/10

Aditya College of Engineering and Technology (affiliated to JNTUK)

Technical Skills

Languages: Java, C, JavaScript (ES6+), TypeScript, SQL

Frontend: React.js, HTML5, CSS3, Bootstrap, Tailwind CSS, EJS

Backend: Node.js, Express.js, REST APIs, MVC Architecture, Spring Boot

Databases: MongoDB, MongoDB Atlas, MySQL, PostgreSQL (Supabase)

Auth & Security: JWT, Passport.js, Session-based Authentication, Bcrypt, Role-Based Access Control

Testing: Jest, JUnit, Mocha, Chai

DevOps & Cloud: Docker, AWS (EC2), Jenkins (CI/CD), Kubernetes, Render

Tools: Git, GitHub, Postman, Cloudinary, Maven, Eclipse

Methodologies: Agile, Scrum

Core CS: Data Structures & Algorithms, Object-Oriented Programming, System Design Basics

Experience

SkillDzire

May 2025 – Jul 2025

Full Stack Developer Intern

- Developed RESTful APIs handling 1000+ requests/day using Node.js and Express.js, improving backend efficiency.
- Built reusable React components, reducing UI rendering issues and improving user experience consistency.
- Integrated frontend and backend systems, reducing API response errors through structured data handling.

SkillDzire

May 2026 – Jul 2026

Generative AI with Large Language Models (LLMs) Intern

- Applied Generative AI and LLM concepts to build AI-powered text generation, conversational AI, and real-world AI applications.
- Developed practical skills in prompt engineering, including zero-shot, few-shot, and role-based prompting, for effective LLM interaction.
- Studied LLM fundamentals, tokenization, transformer architecture, and responsible use of Generative AI.
- Built AI-powered chatbots and content generation systems with API integration and basic deployment workflows.

Projects

BlooderBro – Real-Time Blood Donation Platform

LIVE | GITHUB

- Architected a full-stack MVC platform enabling real-time donor matching, handling 30+ user interactions.
- Designed an automated email alert system notifying donors per request using Nodemailer and SMTP.
- Implemented secure session-based authentication with Bcrypt and MongoDB-backed session storage.
- Built an analytics dashboard using MongoDB aggregation pipelines and Chart.js to track user and request trends.

GuruKul – AI-Powered EdTech Marketplace

GITHUB

- Architected a full-stack MERN marketplace connecting parent, students, tutors, and coaching centers, using TypeScript for end-to-end type safety.
- Engineered an AI-driven "Smart Search" engine integrating Groq AI (LLM) to process natural language and execute complex MongoDB queries for real-time matching.
- Built a scalable file management pipeline using Multer and Cloudinary API for secure PDF/image uploads across student dashboards.
- Implemented Role-Based Access Control (RBAC) with JWT and an event-driven notification system alerting parents on attendance and due dates.

Guardian of Temples – Community Safety & Incident Tracker

LIVE | GITHUB

- Developed a full-stack Next.js 14 (App Router) platform using TypeScript and Tailwind CSS with Server Components for optimal performance.
- Engineered a custom Role-Based Access Control (RBAC) system with Supabase PostgreSQL, enforcing strict Row Level Security (RLS) across multiple user tiers.
- Built a secure moderation CMS and automated incident queue using Server Actions, integrating Cloudinary for media storage and verification.
- Configured Brevo SMTP for secure, URL-based session tokens and transactional emails, successfully scaling user onboarding beyond default API rate limits.

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate • AWS Cloud Practitioner Essentials • Microsoft Security Essentials Professional Certificate • Java Fundamentals – Oracle Academy • Fundamentals of Java Programming – Coursera • Computer Science 101 – edX • Full-Stack Web Developer – LinkedIn Learning (CompTIA Partners) • C Programming – Udemy • Red Hat Introduction to Python (AD141) • Cambridge English Empower B2 Level Course